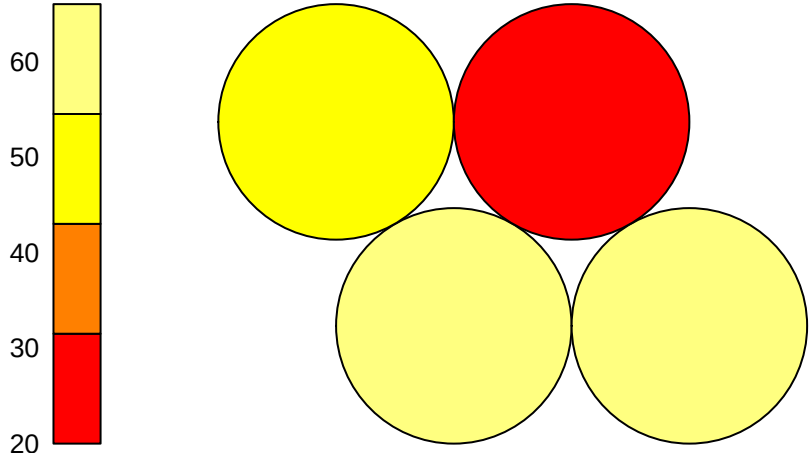
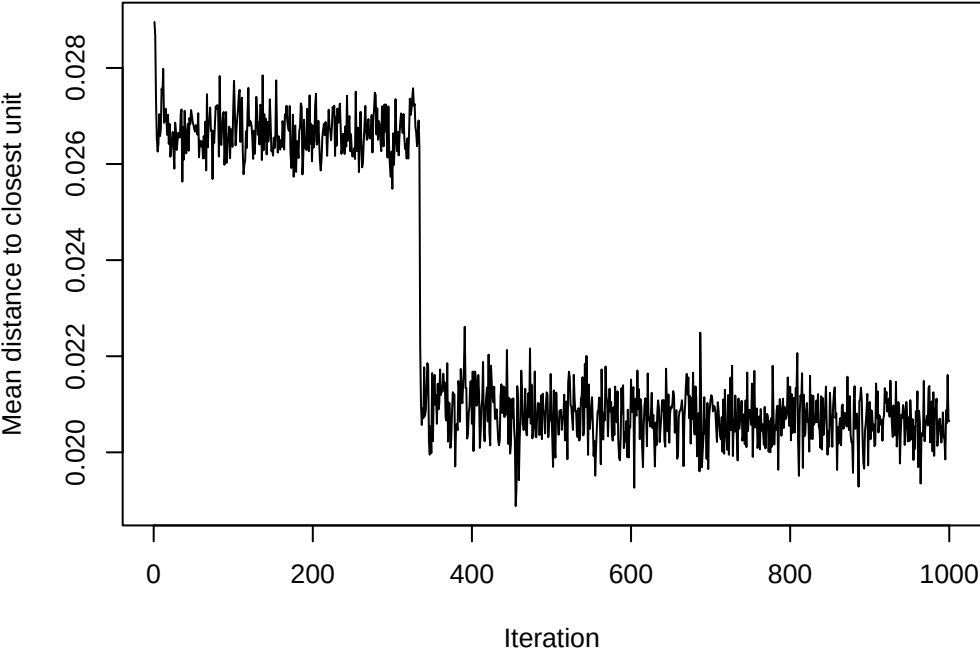


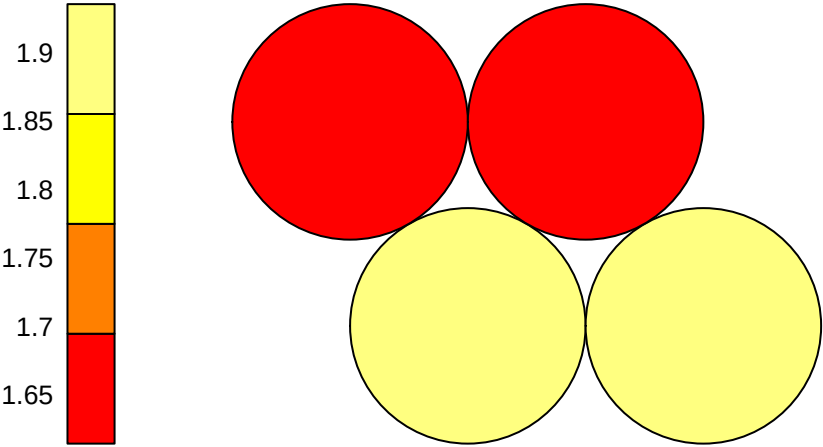
emotions | bubble | Fold 2 – Counts



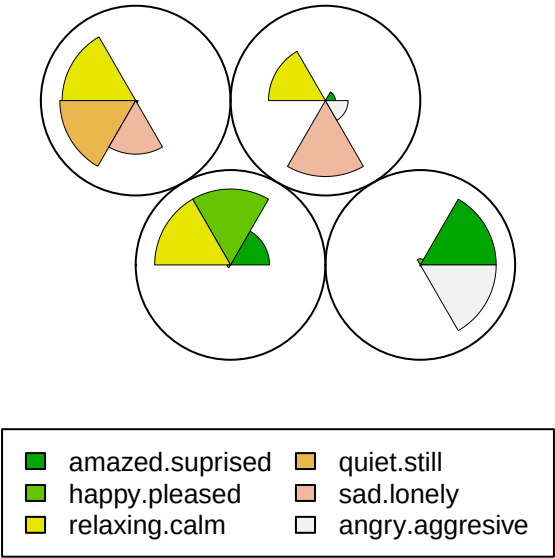
Convergencia do treinamento



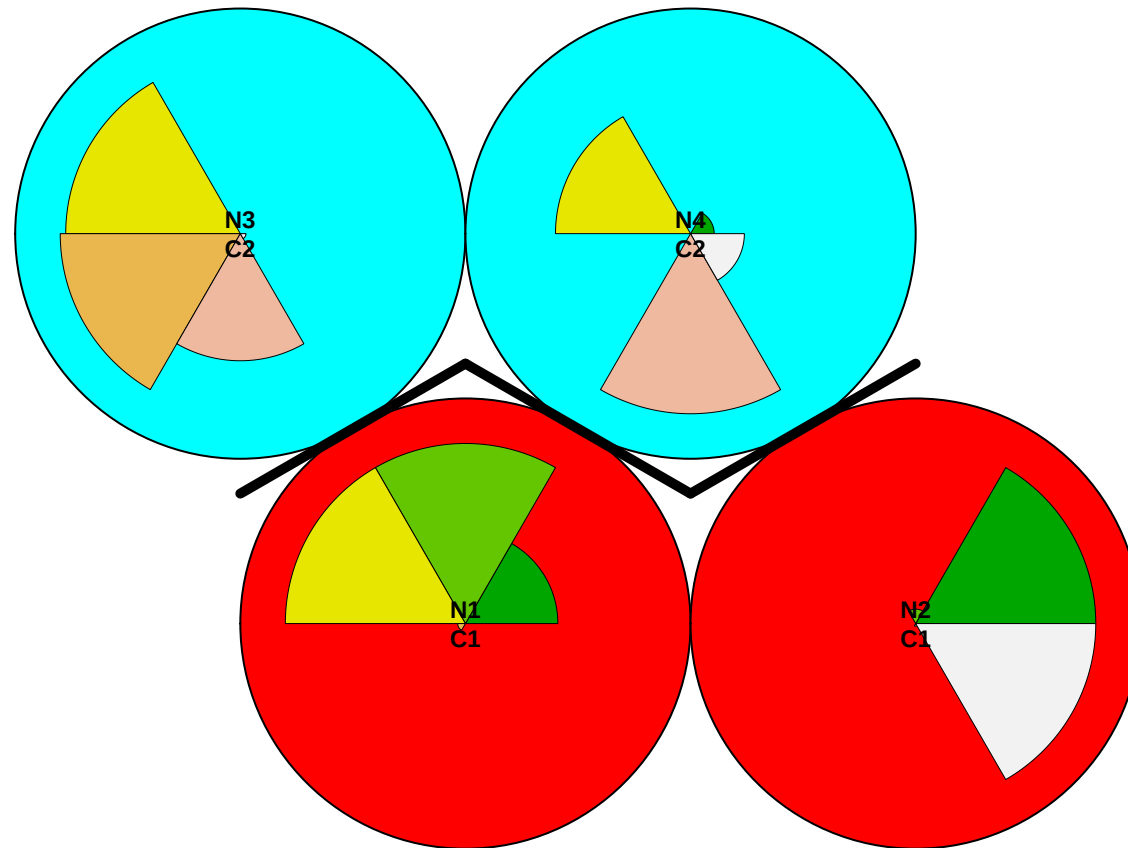
U-Matrix (distancia entre vizinhos)



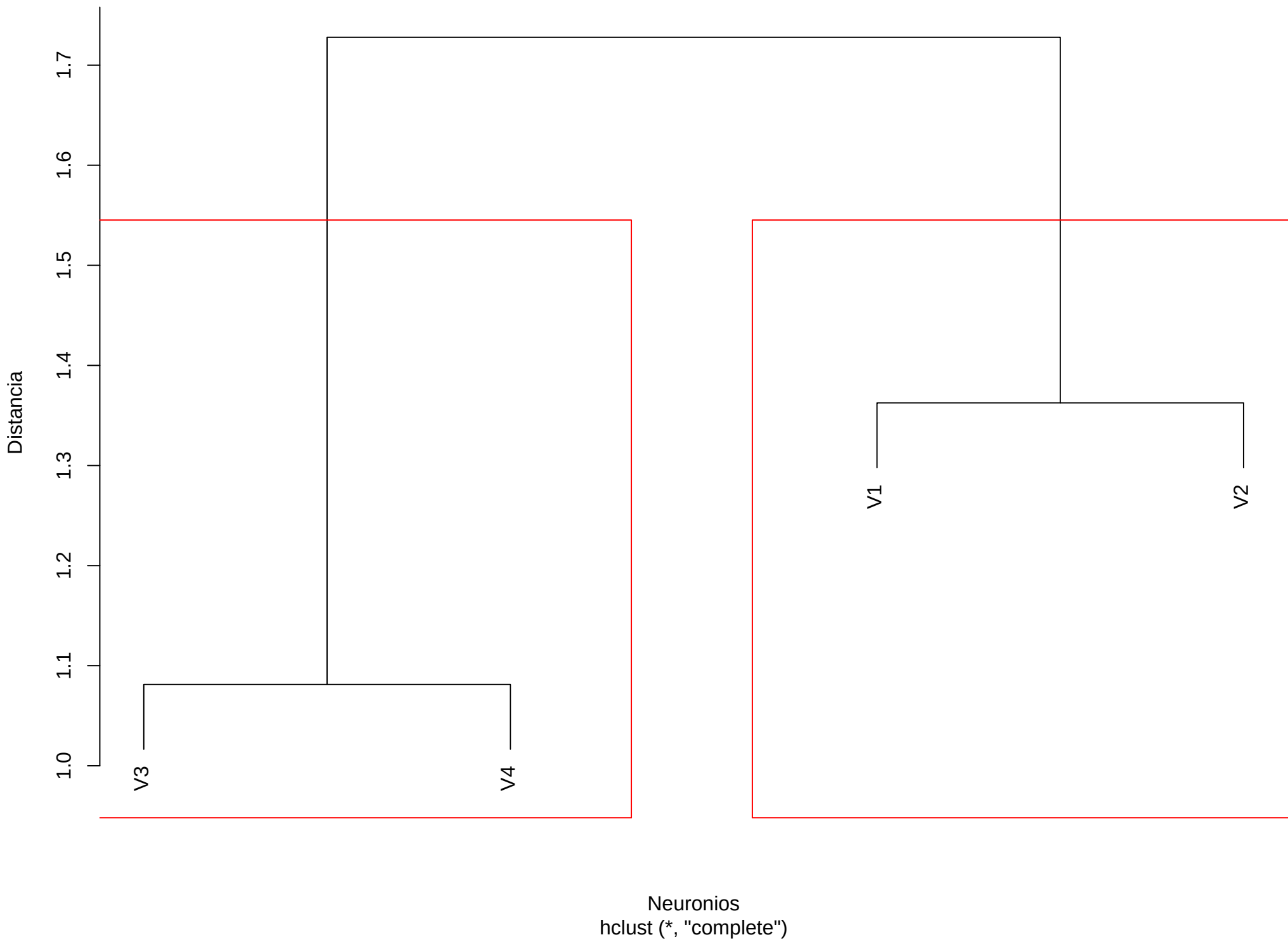
Codebook – Segments



# Clusters (k=2) | emotions | bubble | Fold 2



# Dendrograma (k=2) | emotions | bubble | Fold 2



| <b>Metrica</b>               | <b>Valor</b> | <b>Ideal</b>  |
|------------------------------|--------------|---------------|
| Silhouette medio             | 0.1688       | Próximo de +1 |
| Entropia media dos labelsets | 2.9263       | Próximo de 0  |
| Hellinger (rotulos)          | 0.7069       | Próximo de 1  |
| CCC (pre-corte)              | 0.7315       | Acima de 0.75 |
| AC (pre-corte)               | 0.2928       | Acima de 0.75 |

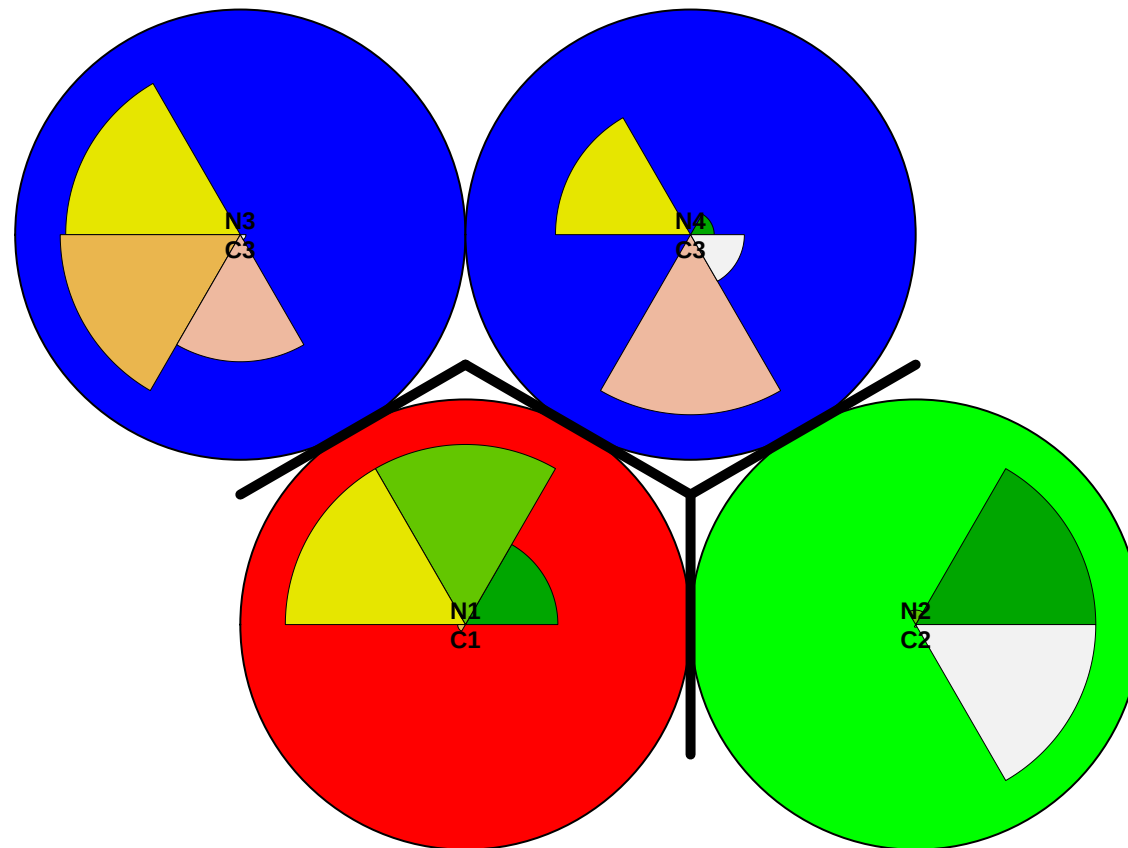
## Frecuencia de rotulos por cluster (k=2) | emotions | bubble | Fold 2

| cluster | n_instancias | amazed.suprised | happy.pleased | relaxing.calm | quiet.still | sad.lonely | angry.aggresive |
|---------|--------------|-----------------|---------------|---------------|-------------|------------|-----------------|
| 1       | 131          | 56              | 55            | 47            | 2           | 1          | 57              |
| 2       | 67           | 2               | 0             | 43            | 47          | 54         | 6               |

| cluster | labelset | frequencia |
|---------|----------|------------|
| 1       | 100001   | 27         |
| 1       | 011000   | 25         |
| 1       | 000001   | 24         |
| 1       | 001000   | 14         |
| 1       | 110000   | 13         |
| 1       | 100000   | 8          |
| 1       | 010000   | 7          |
| 1       | 111000   | 4          |
| 1       | 010001   | 2          |
| 1       | 011100   | 2          |
| 1       | 110001   | 2          |
| 1       | 001001   | 1          |
| 1       | 100011   | 1          |
| 1       | 101000   | 1          |

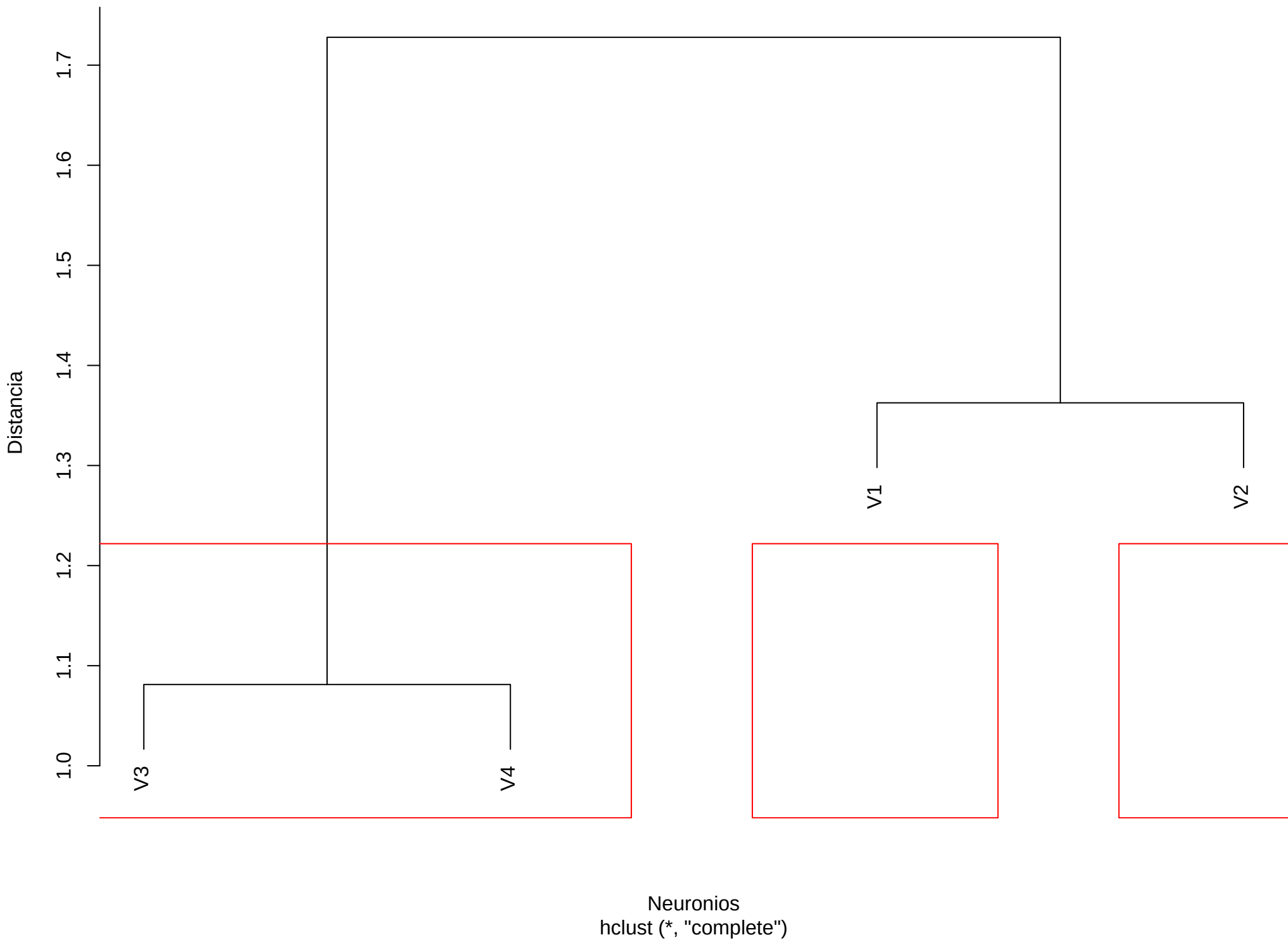
| cluster | labelset | frequencia |
|---------|----------|------------|
| 2       | 001110   | 22         |
| 2       | 000110   | 12         |
| 2       | 001100   | 10         |
| 2       | 001010   | 9          |
| 2       | 000010   | 4          |
| 2       | 000011   | 4          |
| 2       | 000100   | 2          |
| 2       | 100010   | 2          |
| 2       | 001011   | 1          |
| 2       | 001101   | 1          |

# Clusters (k=3) | emotions | bubble | Fold 2





# Dendrograma (k=3) | emotions | bubble | Fold 2



| <b>Metrica</b>               | <b>Valor</b> | <b>Ideal</b>  |
|------------------------------|--------------|---------------|
| Silhouette medio             | 0.1141       | Próximo de +1 |
| Entropia media dos labelsets | 2.3209       | Próximo de 0  |
| Hellinger (rotulos)          | NA (k>2)     | Próximo de 1  |
| CCC (pre-corte)              | 0.7315       | Acima de 0.75 |
| AC (pre-corte)               | 0.2928       | Acima de 0.75 |

## Frecuencia de rotulos por cluster (k=3) | emotions | bubble | Fold 2

| cluster | n_instancias | amazed.suprised | happy.pleased | relaxing.calm | quiet.still | sad.lonely | angry.aggressive |
|---------|--------------|-----------------|---------------|---------------|-------------|------------|------------------|
| 1       | 66           | 18              | 51            | 46            | 2           | 0          | 0                |
| 2       | 65           | 38              | 4             | 1             | 0           | 1          | 57               |
| 3       | 67           | 2               | 0             | 43            | 47          | 54         | 6                |

## Top 15 labelsets ... Cluster 1 (k=3) | emotions | bubble | Fold 2

| cluster | labelset | frecuencia |
|---------|----------|------------|
| 1       | 011000   | 25         |
| 1       | 001000   | 14         |
| 1       | 110000   | 13         |
| 1       | 010000   | 7          |
| 1       | 111000   | 4          |
| 1       | 011100   | 2          |
| 1       | 101000   | 1          |

## Top 15 labelsets ... Cluster 2 (k=3) | emotions | bubble | Fold 2

| cluster | labelset | frecuencia |
|---------|----------|------------|
| 2       | 100001   | 27         |
| 2       | 000001   | 24         |
| 2       | 100000   | 8          |
| 2       | 010001   | 2          |
| 2       | 110001   | 2          |
| 2       | 001001   | 1          |
| 2       | 100011   | 1          |

| cluster | labelset | frequencia |
|---------|----------|------------|
| 3       | 001110   | 22         |
| 3       | 000110   | 12         |
| 3       | 001100   | 10         |
| 3       | 001010   | 9          |
| 3       | 000010   | 4          |
| 3       | 000011   | 4          |
| 3       | 000100   | 2          |
| 3       | 100010   | 2          |
| 3       | 001011   | 1          |
| 3       | 001101   | 1          |